

JEE MAIN CUMULATIVE TEST SERIES



NARAYANA
IIT ACADEMY
INDIA



Sec: JR_*CO-SC(MODEL-A)

Time: 3 Hrs

Date: 30-06-24

Max. Marks: 300

30-06-24_JR.IIT_STAR CO-SC(MODEL-A)_JEE MAIN_CTM-1_SYLLABUS

PHYSICS:

100% CUMULATIVE

Syllabus covered From 15-04-24 TO 29-06-24

CHEMISTRY:

100% CUMULATIVE

Syllabus covered From 15-04-24 TO 29-06-24

MATHEMATICS:

100% CUMULATIVE

Syllabus covered From 15-04-24 TO 29-06-24

THE NARAYANA GROUP

PHYSICS

MAX.MARKS: 100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Assertion: If $y = x^3 + 3x^2$ then there will be a maxima at $x = -2$

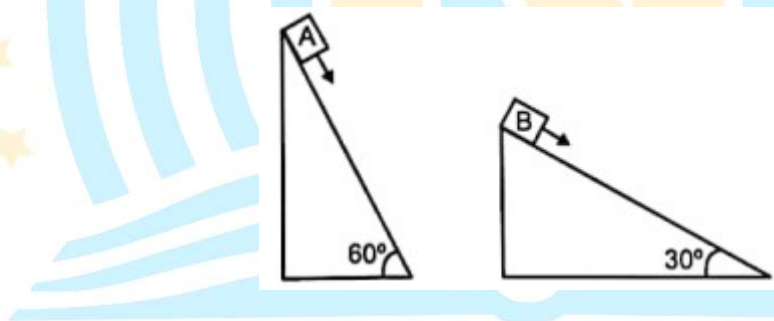
Reason: For this curve $x=0$ is a point of maxima.

A) Both Assertion and Reason are true and the Reason is correct explanation of the Assertion.

B) Both Assertion and Reason are true, but Reason is not correct explanation of the Assertion.

C) Assertion is true, but the Reason is false.

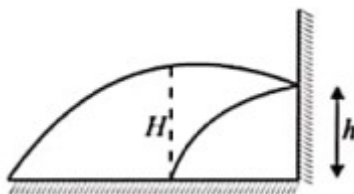
D) Assertion is false, but the Reason is true.
2. Two fixed frictionless inclined planes making an angle 30° and 60° with the horizontal are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B?



- A) 4.9 ms^{-2} in horizontal direction
 - B) 9.8 ms^{-2} in vertical direction
 - C) Zero
 - D) 4.9 ms^{-2} in vertical direction
3. From a tower of height H , a particle is thrown vertically upwards with a speed u . The time taken by the particle to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H , u and n is

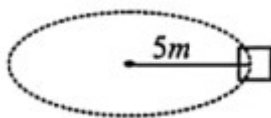
A) $gH = (n-2)^2 u^2$ B) $2gH = nu^2 (n-2)$ C) $gH = (n-2)u^2$ D) $2gH = n^2 u^2$

4. A stone is projected from a horizontal plane. It attains maximum height 'H' & strikes a stationary smooth wall and falls on the ground vertically below the maximum height. Assume the collision to be elastic (rebounds with same speed obeying law of reflection of light) the height of the point on the wall where ball will strike is:

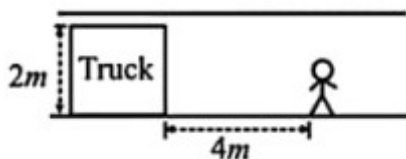


- A) $\frac{H}{2}$ B) $\frac{H}{4}$ C) $\frac{3H}{4}$ D) None of these
5. A block of mass 25 kg rests on a horizontal floor ($\mu = 0.2$). It is attached by a 5m long horizontal rope to a peg fixed on floor. The block is pushed along the ground with an initial velocity of 10 m/s at $t=0$ so that it moves in a circle around the peg. Choose the correct one option.

- A) The magnitude of tangential acceleration is 4ms^{-2} .
 B) Speed of the block at time t is $5 - 2t \text{ m/s}$.
 C) Time when tension in rope becomes zero is 5sec.
 D) All the above are correct.

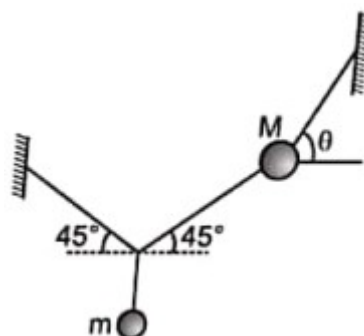


6. A 2m wide truck is moving with a uniform speed of 8 m/s along a straight horizontal road. A pedestrian starts crossing the road at an instant when the truck is 4 m away from him. The minimum constant velocity with which he should run to avoid an accident is

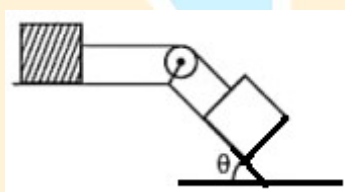


- A) $1.6\sqrt{5} \text{ m/s}$ B) $1.2\sqrt{5} \text{ m/s}$ C) $1.2\sqrt{7} \text{ m/s}$ D) $1.6\sqrt{7} \text{ m/s}$

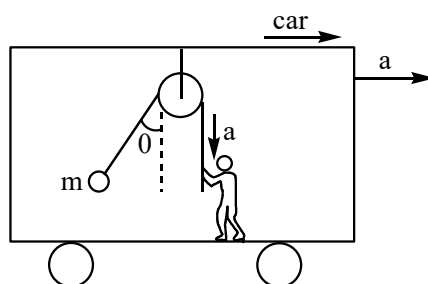
7. Two masses m and M are attached with strings as shown. For the system to be in equilibrium we have



- A) $\tan \theta = 1 + \frac{2M}{m}$ B) $\tan \theta = 1 + \frac{2m}{M}$ C) $\tan \theta = 1 + \frac{M}{2m}$ D) $\tan \theta = 1 + \frac{m}{2M}$
8. Two bodies of identical mass are tied by an ideal string which passes over an ideal pulley. The co-efficient of friction between the bodies and the plane is μ . The minimum value of θ for which the system starts moving is

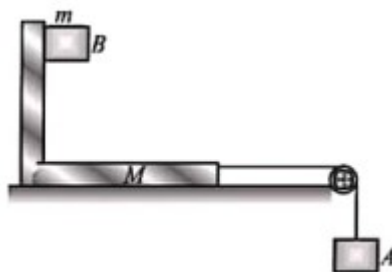


- A) $\cos^{-1}\left(\frac{\mu^2 - 1}{\mu^2 + 1}\right)$ B) $\cos^{-1}\left(\frac{\mu^2 + 1}{1 - \mu^2}\right)$ C) $\cos^{-1}\left(\frac{2\mu}{1 + \mu^2}\right)$ D) $\theta = \cos^{-1}\left(\frac{1 - \mu^2}{1 + \mu^2}\right)$
9. A bob is hanging over a pulley inside a car through a string. The second end of the string is in the hand of a person standing in the car. The car is moving with constant acceleration a directed horizontally as shown in figure. Other end of the string is pulled with constant acceleration a vertically down. The tension in the string is equal to

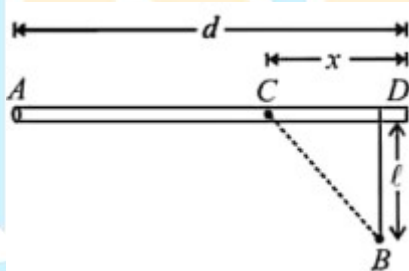


- A) $m\sqrt{g^2 + a^2}$ B) $m\sqrt{g^2 + a^2} - ma$ C) $m\sqrt{g^2 + a^2} + ma$ D) $m(g + a)$

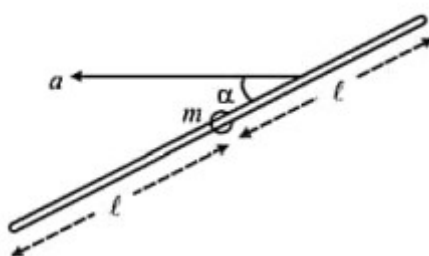
10. The figure shows an L shaped body of mass M placed on smooth horizontal surface. The block A is connected to the body by means of an inextensible string, which is passing over a smooth pulley of negligible mass. Another block B of mass m is placed against a vertical wall of the body. Find the Minimum value of the mass of block A so that block B remains stationary relative to the wall. Coefficient of friction between the block B and the vertical wall is μ is large enough.



- A) $\frac{\mu(M+m)}{1-\mu}$ B) $\frac{\mu M}{1+\mu}$ C) $\frac{M+m}{\mu-1}$ D) $\frac{\mu m}{\mu+1}$
11. From point 'A' located on a highway one has to get by car as soon as possible to point 'B' located in the field at a distance ' ℓ ' from the highway. It is known that the car moves in the field η times slower than on the highway? At what distance from point D one must turn off the highway?



- A) $\frac{\eta \ell}{\sqrt{\eta^2-1}}$ B) $\frac{\eta \ell}{\sqrt{\eta^2+1}}$ C) $\frac{\ell}{\sqrt{\eta^2+1}}$ D) $\frac{\ell}{\sqrt{\eta^2-1}}$
12. A bead of mass ' m ' is fitted onto a rod with a length of 2ℓ , and can move on it with friction having the coefficient of friction μ . At the initial moment the bead is in the middle of the rod. The rod moves translationally in a horizontal plane with an acceleration ' a ' in the direction forming an angle α with the rod. The time when the bead will leave the rod is: (Neglect the weight of the bead).



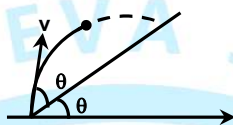
A) $\sqrt{\frac{2l}{a(\cos \alpha - \mu \sin \alpha)}}$

B) $\sqrt{\frac{2l}{a(\sin \alpha - \mu \cos \alpha)}}$

C) $\sqrt{\frac{4l}{a(\sin \alpha - \mu \cos \alpha)}}$

D) None of these

13. A particle is projected with a velocity u at an angle θ with an inclined plane which makes an angle $\theta < 45^\circ$ with the horizontal. Calculate the radius of curvature of the path of projectile when velocity of projectile becomes parallel to the plane.



A) $\frac{u^2 \cos^2 2\theta}{g \cos^3 \theta}$

B) $\frac{u^2 \sin^2 2\theta}{g \cos^3 \theta}$

C) $\frac{u^2 \cos 2\theta}{g \sin^3 \theta}$

D) $\frac{u^2 \cos 2\theta}{g \cos^2 \theta}$

14. The horizontal range and maximum height attained by a projectile are R and H respectively. If a constant horizontal acceleration $a = g/4$ is imparted to the projectile due to wind then its horizontal range and maximum height will be

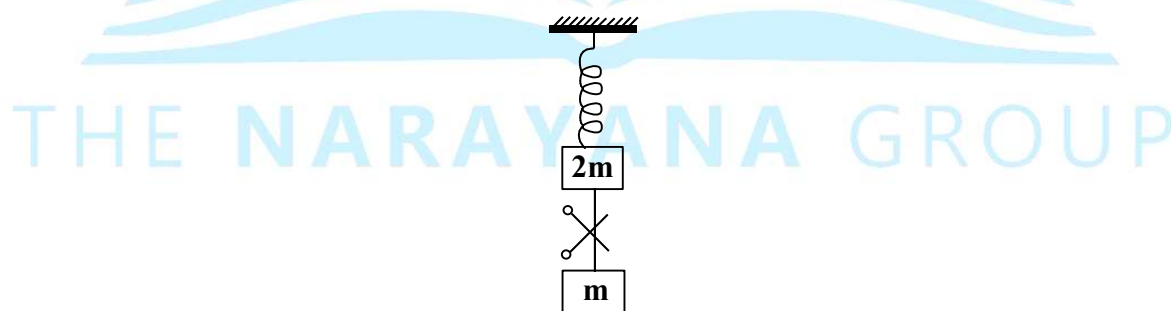
A) $(R+H), H/2$

B) $(R+H/2), 2H$

C) $(R+2H), H$

D) $(R+H), H$

15. System shown in fig is in equilibrium and at rest. The spring and string are massless, now the string is cut. The acceleration of mass $2m$ and m just after the string is cut will be :



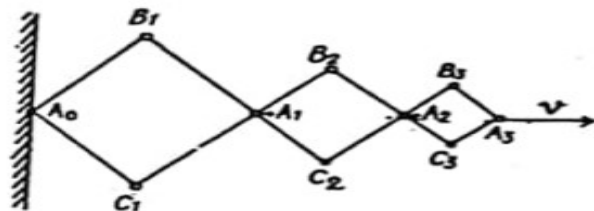
A) $g/2$ upwards, g downwards

B) g upwards, $g/2$ downwards

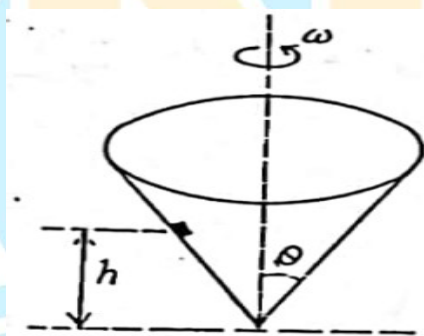
C) g upwards, $2g$ downwards

D) $2g$ upwards, g downwards

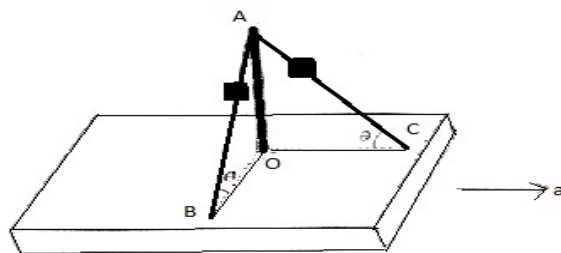
16. A hinged construction consists of three rhombi with the ratio of sides 3:2:1 (Fig.). Vertex A_3 moves in the horizontal direction at a velocity v . Determine the velocities of vertices A_1, A_2 and B_2 at the instant when the angles of the construction are 90° .



- A) $V_{A1} = \frac{V}{12}$ B) $V_{A2} = \frac{5V}{6}$
 C) $V_{B2} = \frac{\sqrt{15}V}{6}$ D) All the above are WRONG
17. A small block of mass m is placed inside a hollow cone rotating about a vertical axis with angular velocity ω as in figure. The semi-vertex angle of the cone is θ and the coefficient of friction between the cone and the block is μ . If the block is to remain at a constant height h above the apex of the cone, what is the maximum value of ω ?

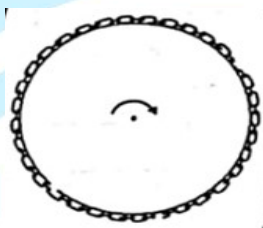


- A) $\omega_{\max} = \sqrt{\frac{g}{h \tan \theta} \left(\frac{\tan \theta - \mu}{\tan \theta + \mu} \right)}$ B) $\omega_{\max} = \sqrt{\frac{g}{h \cot \theta} \left(\frac{1 - \mu \cot \theta}{\cot \theta + \mu} \right)}$
 C) $\omega_{\max} = \sqrt{\frac{g}{h \cot \theta} \left(\frac{1 + \mu \cot \theta}{\cot \theta - \mu} \right)}$ D) $\omega_{\max} = \sqrt{\frac{g}{h \tan \theta} \left(\frac{1 + \mu \tan \theta}{\tan \theta - \mu} \right)}$
18. A horizontal wooden block has a fixed rod OA standing on it. From A two wires AB and BC are fixed to the block. ΔOAB is perpendicular to ΔOAC . One wire is perfectly smooth and one wire is rough. There are two identical beads on the two wires. Now the wooden plank is moved with horizontal acceleration perpendicular to the line OB so that Both beads do not slide on the wires. The coefficient of friction between the rough wire and bead is..



- A) $\frac{\cos \theta}{\sqrt{\cos^2 \theta + \cot^2 \theta}}$ B) $\frac{\sin \theta}{\sqrt{\cos^2 \theta + \tan^2 \theta}}$ C) $\frac{\sin \theta}{\sqrt{\sin^2 \theta + \cot^2 \theta}}$ D) $\frac{\cos \theta}{\sqrt{\sin^2 \theta + \tan^2 \theta}}$

19. A metallic chain of mass m with a length ℓ and whose ends are joined together is fitted onto a wooden light disc as shown in the figure. The disc rotates with a speed of n revolutions per second. Find the tension of the chain T .



- A) mn^2l B) $\frac{mn^2l}{2\pi}$ C) $2\pi mn^2l$ D) $\frac{mn^2l}{4\pi}$
20. A person of mass 60 kg is in a lift. The change in the apparent weight of the person, when the lift moves up with an acceleration of 2ms^{-2} and then down with an acceleration of 2ms^{-2} , is (take $g = 10 \text{ m/sec}^2$)
- A) 120 N B) 240 N C) 480 N D) 720 N

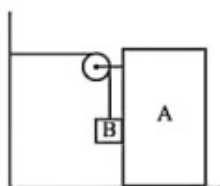
SECTION-II (NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only**. Have to Answer any 5 only out of 10 questions and question will be evaluated according to the following marking scheme:

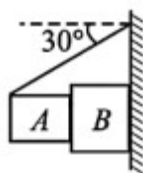
Marking scheme: +4 for correct answer, -1 in all other cases.

21. A 20 kg block B is suspended from a cord attached to a 40 kg cart A. If the acceleration of the block B shown in figure immediately after the system is released from rest is

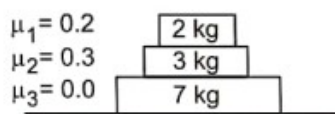
$\frac{\sqrt{X}g}{Y}$, where X, Y are least possible integers. Find the value of X+Y. (neglect friction)



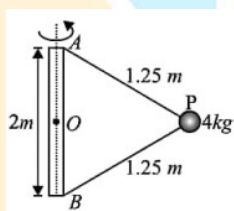
22. Two blocks A and B of mass m 10 kg and 20 kg respectively are placed as shown in figure. Coefficient of friction between all the surfaces is 0.2. Then find tension in string. (in N) ($g = 10 \text{ m/s}^2$)



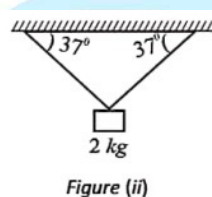
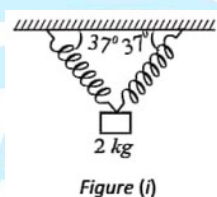
23. The acceleration of 2Kg block shown in the figure is A when a horizontal force of 10N is applied on 2 kg block. The acceleration of 3Kg block shown in the figure is B when a horizontal force of 10N is applied on 3 kg block. If the value of AB is $\frac{x}{2} \text{ m/s}^2$, find the value of x. (Take $g = 10 \text{ m/s}^2$)



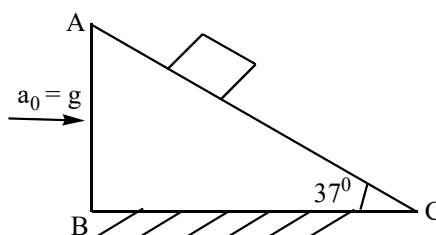
24. The 4 kg block in the figure is attached to the vertical rod by means of two strings. When the system rotates about the axis of the rod as indicated in figure in order that the tension in upper string is 60 N, What is tension in the lower string. (in N)?



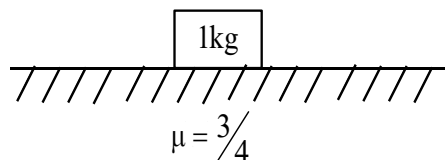
25. The blocks are of mass 2kg shown is in equilibrium. At $t = 0$ right spring in fig (i) and right string in fig (ii) breaks. If the ratio acceleration of blocks in fig (i) to fig(ii) just after break is $\frac{x}{24}$, find the value of x.



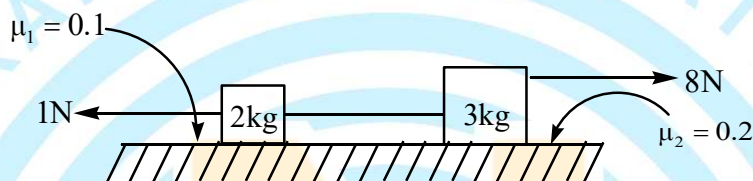
26. A block is placed on an inclined plane moving towards right horizontally with acceleration $a_0 = g$, The length of plane AC = 1m, friction is absent everywhere. The time taken by the block to reach from C to A is (Take $g = 10 \text{ m/s}^2$) (in sec)



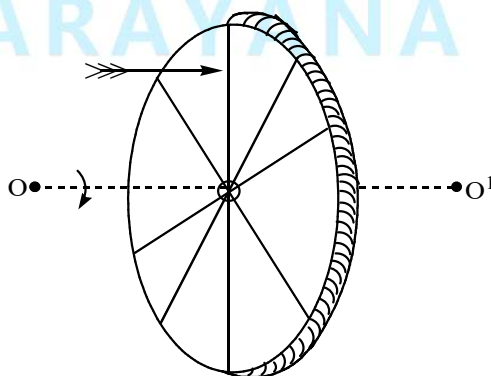
27. A block of mass 1kg is placed on the rough horizontal surface as shown in the fig. The coefficient of friction between them ($\mu = \frac{3}{4}$) the minimum force (in N) required to move the block is ... ($g = 10\text{m/sec}^2$)



28. In the shown arrangement, if T be the tension in the string connecting the blocks then the value of T (in N) is ($g = 10\text{m/sec}^2$)



29. A point traversed $\frac{3}{4}$ th of the circle of radius R in time t. If the magnitude of the average velocity of the particle in this time interval is $\frac{\sqrt{X}R}{t}$, find the value of X.
30. A wheel has eight equally spaced, very thin spokes of length 30 cm. It is rotating about axis OO^1 which is perpendicular to its plane and passing through center, with 2.5 rps. We want to pass a 20 cm long arrow parallel to the axis and without hitting any spoke. Find the minimum velocity of the arrow(in m/s)? [Assumed as initially the arrow is just ready to pass]



CHEMISTRY

MAX.MARKS: 100

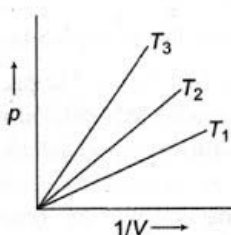
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Oxidation state of three Bromine atoms in Br_3O_8 compound is / are
- A) +6, +6, +6 B) +6, +4, +6 C) +8, 0, +8 D) +6, 0, +6
32. 6.02×10^{20} molecules of urea are present in 100ml of its solution the concentration of solution is
- A) 0.001 mol / lit B) 0.01 mol / lit C) 0.02 mol / lit D) 0.1 mol / lit
33. 8g of NaOH is dissolved in 18g of H_2O . Mole fraction of NaOH in solution and molality (in mol kg^{-1}) of the solution respectively are
- A) 0.2, 11.11 B) 0.167, 22.20 C) 0.2, 22.20 D) 0.167, 11.11
34. Two moles of an ideal gas is heated at constant pressure of one atmosphere from 27°C to 127°C . If $C_{v,m} = 20 + 10^{-2} T \text{ JK}^{-1} \text{ mol}^{-1}$, then q and ΔU for the process are respectively
- A) 6362.8 J, 4700 J B) 30372.2 J, 4700 J
C) 7062.8 J, 5400 J D) 3181.4 J, 2350 J
35. Which of the following reactions is not correct according to the law of conservation of mass?
- A) $2\text{Mg}(s) + \text{O}_2(g) \longrightarrow 2\text{MgO}(s)$ B) $\text{C}_3\text{H}_8(g) + \text{O}_2(g) \longrightarrow \text{CO}_2(g) + \text{H}_2\text{O}(g)$
C) $\text{P}_4(s) + 5\text{O}_2(g) \longrightarrow \text{P}_4\text{O}_{10}(s)$ D) $\text{CH}_4(g) + 2\text{O}_2(g) \longrightarrow \text{CO}_2(g) + 2\text{H}_2\text{O}(g)$
36. Consider a graph plotted between p and $\frac{1}{V}$ as shown below

The correct order of temperature T_1, T_2, T_3 will be



- 1) $T_3 > T_2 > T_1$ B) $T_1 > T_2 > T_3$ C) $T_1 = T_2 = T_3$ D) $T_1 > T_2 = T_3$

- Page. No. 12

42. The entropy change associated with the conversion of 1 kg of ice at 273 K to water vapours at 383K is__ (Specific heat of water liquid and water vapour are $4.2 \text{ kJ K}^{-1} \text{ kg}^{-1}$ and $2.0 \text{ kJ K}^{-1} \text{ kg}^{-1}$; heat of liquid fusion and vapourisation of water are 334 kJ kg^{-1} and 2491 kJ kg^{-1} respectively).

($\log 273 = 2.436$, $\log 373 = 2.572$, $\log 383 = 2.583$)

A) $9.26 \text{ kJ kg}^{-1} \text{ K}^{-1}$ B) $8.49 \text{ kJ kg}^{-1} \text{ K}^{-1}$

C) $7.90 \text{ kJ kg}^{-1} \text{ K}^{-1}$ D) $2.64 \text{ kJ kg}^{-1} \text{ K}^{-1}$

43. The enthalpy of combustion of methane, graphite and dihydrogen at 298 K are, $-890.3 \text{ kJ mol}^{-1}$, $-393.5 \text{ kJ mol}^{-1}$ and $-285.8 \text{ kJ mol}^{-1}$ respectively. Enthalpy of formation of $\text{CH}_4(\text{g})$ will be

A) $-74.8 \text{ kJ mol}^{-1}$ B) $-52.27 \text{ kJ mol}^{-1}$ C) $+74.8 \text{ kJ mol}^{-1}$ D) $+52.26 \text{ kJ mol}^{-1}$

44. **Assertion (A) :** U is a state function.

Reason (R): T is an intensive property.

A) Both A and R are correct; R is the correct explanation of A.

B) Both A and R are correct; R is not the correct explanation of A

C) A is correct; R is incorrect

D) A is incorrect; R is correct

45. Match the Column I with Column II and mark the option given in the codes below

	Column -I		Column -II
A	$\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}$	1.	$\Delta_{\text{sol}} H^\circ$
B	$\text{H}_2(\text{g}) \longrightarrow 2\text{H}(\text{g})$	2.	$\Delta_{\text{lattice}} H^\circ$
C	$\text{NaCl}(\text{s}) \longrightarrow \text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$	3.	$\Delta_c H^\circ$
D	$\text{NaCl}(\text{s}) \longrightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	4	$\Delta_{\text{bond}} H^\circ$

A) A-4; B-3; C-1; D-2

B) A-2; B-1; C-4; D-3

C) A-1; B-2; C-3; D-4

D) A-3; B-4; C-2; D-1

46. Match the following parameters with description for spontaneity and choose the correct codes given below.

	$(\Delta_r H^\circ, \Delta_r S^\circ, \Delta_r G^\circ \text{ Parameters})$		Description
A	+ - +	1.	Non-spontaneous at high temperature
B	- - +	2.	Spontaneous at all temperatures
C	- + -	3.	Non-spontaneous at all temperatures

A) A-1; B-3; C-2

B) A-3; B-2; C-1

C) A-1; B-2; C-3

D) A-3; B-1; C-2

47. 1.1 mole of A is mixed with 2.2 mole of B and the mixture is kept in a 1 L flask till the equilibrium, $A + 2B \rightleftharpoons 2C + D$ is reached. At equilibrium, 0.2 mole of C is formed. The equilibrium constant would be

A) 0.002

B) 0.004

C) 0.001

D) 0.003

48. For a reaction taking place in 3 steps, the rate constants are K_1, K_2 and K_3 . The activation energy values for the 1st, 2nd and 3rd stages are 40, 50 and 60 KJ / mole respectively, then overall activation energy for $K = \frac{K_1 K_2}{K_3}$ in KJ / mol is

A) 40

B) 30

C) 150

D) 70

49. 5.1g NH_4HS is introduced in 3.0 L evacuated flask at $327^\circ C$ 30% of the solid NH_4HS decomposed to NH_3 and H_2S as gases. The K_p of the reaction at $327^\circ C$ is ($R = 0.082 \text{ atm mol}^{-1} K^{-1}$, molar mass of $S = 32 g \text{ mol}^{-1}$ molar mass of $N = 14 g \text{ mol}^{-1}$)

A) $0.242 \times 10^{-4} \text{ atm}^2$ B) 0.242 atm^2 C) $4.9 \times 10^{-3} \text{ atm}^2$ D) $1 \times 10^{-4} \text{ atm}^2$

50. The standard Gibbs energy change at 300 K, for the reaction, $2A \rightleftharpoons B + C$ is 2494.2J. At a given time, the composition of the reaction mixture is

$[A] = 1/2$, $(B) = 2$ and $[C] = 1/2$. The reaction proceeds in the ($R = 8.314 \text{ J/K mol}$, $e = 2.718$)

A) forward direction because $Q > K_C$ B) reverse direction because $Q > K_C$ C) forward direction because $Q < K_C$ D) reverse direction because $Q < K_C$

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

51. Equal volumes of two acids having H^+ ion concentrations $10^{-4}N$ and $10^{-6}N$ are mixed. The pH of the resulting solution is _____
52. If the maximum concentration of $PbCl_2$ in water is $0.01 M$ at $25^\circ C$, its maximum concentration in $0.1M NaCl$ will be 4×10^{-x} , the value of x is _____
53. Given $Ag(NH_3)_2^+ \rightleftharpoons Ag^+ + 2NH_3$;
 $K_C = 6.2 \times 10^{-8}$ and K_{sp} of $AgCl$ is 1.8×10^{-10} at $298 K$. The concentration of the complex in $1M$ aqueous ammonia, if ammonia is added to a water solution containing excess of $AgCl(s)$ is $x \times 10^{-4}$, the value of x is _____
54. pH of saturated solution of $Ba(OH)_2$ is 12. The value of solubility product of K_{sp} of $Ba(OH)_2$ is $x \times 10^{-7}$, the value of x is _____
55. What is the minimum pH of a solution having concentration $0.10 M$ of Mg^{2+} , from which $Mg(OH)_2$ will not precipitate? $K_{sp} Mg(OH)_2 = 1.2 \times 10^{-11}$
56. 90% of a 1st order reaction completed in 30 minutes. The time required to complete 99.9% reaction is 'x' times to the time taken for the completion of 90% of the reaction, the value of 'x' is _____
57.

	[A]	[B]	rate
1)	0.1	0.2	4×10^{-2}
2)	0.1	0.4	8×10^{-2}
3)	0.2	0.4	3.2×10^{-1}

Over all order of the reaction is _____
58. The concentration of a reaction changes from $0.4M$ to $0.2M$ in 1 hour, and the concentration changes from $1M$ to $0.25M$ in 2 hours the order of the reaction would be _____
59. The half life of the 1st order reaction

$$C_2H_4O \xrightarrow{(g)} CH_4 + CO \text{ is if } (g) \quad (g)$$

The initial pressure of C_2H_4O is $80 mm$ and the total pressure at the end of 20 minutes is $120mm$
60. If the half lives of the first order reaction at $350K$ and $300K$ are 2 and 20 seconds respectively. The activation energy of the reaction in kJ is nearly rounded off to x , the value of x is _____

MATHEMATICS

MAX.MARKS: 100

SECTION – I

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

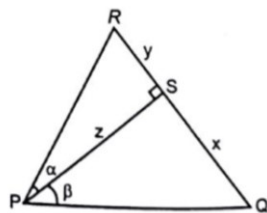
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. If $\sin \theta + \operatorname{cosec} \theta = 2$, then the value of $\sin^8 \theta + \operatorname{cosec}^8 \theta$ is equal to
 A) 2 B) 2^4 C) 2^8 D) more than 2^8
62. If in a triangle ABC, a, b, c are the length of the sides BC, CA, AB respectively and they are in A.P. then it is necessary that
 A) $\frac{2}{3} < \frac{b}{c} < 2$ B) $\frac{1}{3} < \frac{b}{c} < \frac{2}{3}$ C) $\frac{1}{4} < \frac{b}{c} < \frac{1}{3}$ D) $\frac{1}{3} < \frac{b}{c} < \frac{4}{3}$
63. If the equation $|x^2 + bx + c| = k$ has four distinct real roots, then
 A) $b^2 - 4c > 0$ and $0 < k < \frac{b^2 - 4c}{4}$ B) $b^2 - 4c < 0$ and $0 < k < \frac{4c - b^2}{4}$
 C) $b^2 - 4c > 0$ and $k > \frac{b^2 - 4c}{4}$ D) $b^2 - 4c < 0$ and $k > \frac{b^2 - 4c}{4}$
64. The product $\prod_{k=1}^n \left(1 - \tan^2 \frac{2^k \pi}{2^n + 1} \right)$ will be equal to
 A) 2^n B) -2^n C) 2^{-n} D) $(-2)^n$
65. Suppose $a_1, a_2, a_3, \dots, a_{2012}$ are integers arranged on a circle. Each number is equal to the A.M of its two adjacent numbers. If the sum of all even index (with even suffixes) numbers is 3018 then the sum of all the numbers is
 A) 12072 B) 9054 C) 4027 D) 6036
66. If $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots + \frac{1}{199} - \frac{1}{200} = \frac{1}{101} + \frac{1}{102} + \frac{1}{103} + \dots + \frac{1}{m}$ then the numerical positive integer $\frac{m}{50}$ should be equal to
 A) 0 B) 3 C) 4 D) 1

67. Sum of values of x and y satisfying the equation $3^x - 4^y = 77$; $3^{x/2} - 2^y = 7$ is
- A) 2 B) 3 C) 4 D) 5
68. If S_1, S_2 and S_3 are the sums of first n natural numbers, their squares and their cubes respectively, then $\frac{S_1^4 S_2^2 - S_2^2 S_3^2}{S_1^2 + S_3^2} =$
- A) 4 B) 2 C) 1 D) 0
69. $f(x) = x^6 - 12x^5 + ax^4 + bx^3 + cx^2 + dx + 64 = 0$ has all positive roots then $a + b + c + d =$
- A) -52 B) -50 C) -48 D) -40
70. If $a + b + c = 6$ then maximum value of $\sqrt{4a+1} + \sqrt{4b+1} + \sqrt{4c+1}$ is
- A) 6 B) 9 C) 12 D) 18
71. If $p, q, r, s \in R$ then the equation $(x^2 + px + 3q)(-x^2 + rx + q)(-x^2 + sx - 2q) = 0$ must have
- A) 6 real roots B) at least two real roots
- C) 4 real & 2 imaginary roots D) 2 real & 4 imaginary roots
72. Two consecutive numbers $k, k + 1$ are removed from natural numbers $1, 2, 3, \dots, n$ such that their arithmetic mean reduces by 1, then $(n - k)$ is equal to
- A) 1 B) 2 C) 3 D) 4
73. If $\sum_{n=1}^{360} \left(\frac{1}{n\sqrt{n+1} + (n+1)\sqrt{n}} \right) = \frac{p}{q}$ (where p and q relatively prime integers), then $|p - q|$ is
- A) 0 B) 1 C) 2 D) 3
74. If $4\cos 36^\circ + \cot\left(7\frac{1}{2}\right)^0 = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3} + \sqrt{n_4} + \sqrt{n_5} + \sqrt{n_6}$ (where $n_1, n_2, n_3, n_4, n_5, n_6$ natural numbers), then the value of $\sum_{i=1}^6 (n_i)^2$
- A) 19 B) 91 C) 81 D) 101

75. If $\log_3 \sin x - \log_3 \cos x - \log_3 (1 - \tan x) - \log_3 (1 + \tan x) = -1$, then $\tan 2x$ is equal to (wherever defined)
- A) -2 B) $\frac{3}{2}$ C) $\frac{2}{3}$ D) 6
76. Let $f(x) = x^2 + px + q$ and $g(x) = x^2 + rx + s$ where real roots of $f(x) = 0$ are α, β and real roots of $g(x) = 0$ are $\alpha + \delta, \beta + \delta$ such that minimum value of $f(x)$ be $-\frac{1}{4}$ and least value of $g(x)$ occurs at $x = \frac{7}{2}$, then correct option is _____
- A) Least value of $g(x)$ is $\frac{1}{2}$
- B) Least value of $g(x)$ is $-\frac{1}{2}$
- C) Roots of the equation $g(x) = 0$ are $3, 4$
- D) Roots of the equation $g(x) = 0$ are $\frac{3}{2}, \frac{7}{2}$
77. The sum of the infinite series $1 + \left(1 + \frac{1}{2}\right)\left(\frac{1}{3}\right) + \left(1 + \frac{1}{2} + \frac{1}{2^2}\right)\left(\frac{1}{3^2}\right) + \dots \infty$ is S , then $5S$ is
- A) 9 B) 6 C) 8 D) 5
78. $\sum_{r=1}^8 \tan(rA) \tan((r+1)A)$ where $A = 36^\circ$ is
- A) $-10 - \tan A$ B) $-10 + \tan A$ C) -10 D) -9
79. If $u = \sqrt{a^2 \cos^2 \theta + b^2 \sin^2 \theta} + \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$, then the difference between the maximum and minimum values of u^2 given by
- A) $2(a^2 + b^2)$ B) $2\sqrt{a^2 + b^2}$ C) $(a+b)^2$ D) $(a-b)^2$

80. In a ΔPQR (as shown in figure) if $x : y : z = 2 : 3 : 6$, then the value of $\angle QPR$ is



A) $\frac{\pi}{6}$

B) $\frac{\pi}{4}$

C) $\frac{\pi}{3}$

D) $\frac{\pi}{2}$

SECTION-II (NUMERICAL VALUE ANSWER TYPE)

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81. If $a > 0, b > 0, c > 0$ and the minimum value of $a(b^2 + c^2) + b(c^2 + a^2) + c(a^2 + b^2)$ is λabc then λ is ____.

82. The number of solutions of $|\lfloor x \rfloor - 2x| = 4$, where $\lfloor x \rfloor$ is the greatest integer $\leq x$, is

83. If $\sin \alpha, \sin \beta, \sin \gamma$ are in A.P. and $\cos \alpha, \cos \beta, \cos \gamma$ are in G.P. then

$$\left| \frac{\cos^2 \alpha + \cos^2 \gamma - 4 \cos \alpha \cos \gamma}{1 - \sin \alpha \sin \gamma} \right| \text{ equals}$$

84. If $a \neq 0, b \neq 0, ax^2 + bx + c = 0, bx^2 + cx + a = 0$ have a common root then the value of $2^{\frac{a^2}{bc}} \cdot 2^{\frac{b^2}{ca}} \cdot 2^{\frac{c^2}{ab}}$ is equal to:

85. The sum of the roots of the equation $9^{\log_3(\log_2^x)} = \log_2^x - (\log_2^x)^2 + 1$ is ____

86. The number of real roots of the equation $9 \left(\sqrt{\frac{x}{2-x}} - 8 \left(\sqrt{\frac{2-x}{x}} \right) \right) + 1 = 0$ is

87. If $n(A) = 70, n(B) = 80, n(C) = 90$, then find the minimum value of $n(A \cap B \cap C)$ where $n(U) = 100$, U being the universal set.

88. If $\alpha, \beta, \gamma, \delta$ be four roots of the equation $x^4 - 10x^3 + 37x^2 - 60x + 32 = 0$. If $\frac{1}{\alpha^2 - 5\alpha + 10} + \frac{1}{\beta^2 - 5\beta + 10} + \frac{1}{\gamma^2 - 5\gamma + 10} + \frac{1}{\delta^2 - 5\delta + 10} = \frac{p}{q}$, where p and q are co-prime numbers then $p + q$ is
89. If x_1, x_2, x_3, x_4 and x_5 be 5 positive real numbers such that $x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5 = 15$ and $x_1 \cdot x_2^2 \cdot x_3^3 \cdot x_4^4 \cdot x_5^5 = 1$ then $x_1 + x_2 + x_3 + x_4 + x_5$ equals
90. Let a, b, c be positive real numbers, such that, $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \leq 1$, now if the least value of $(1+a^2)(1+b^2)(1+c^2)$ is λ , then λ equals

